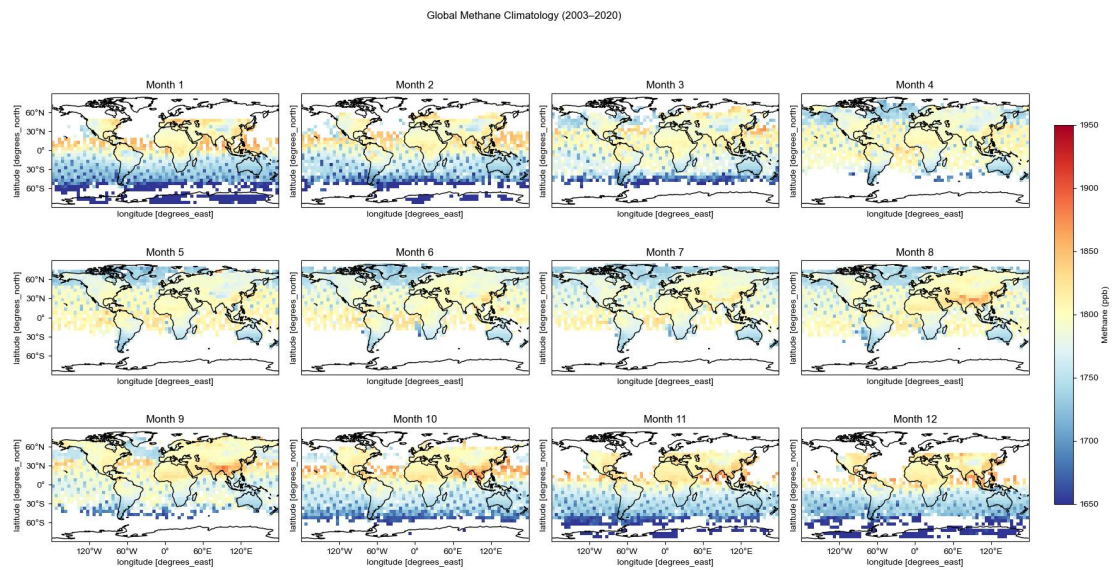


Assignment 3

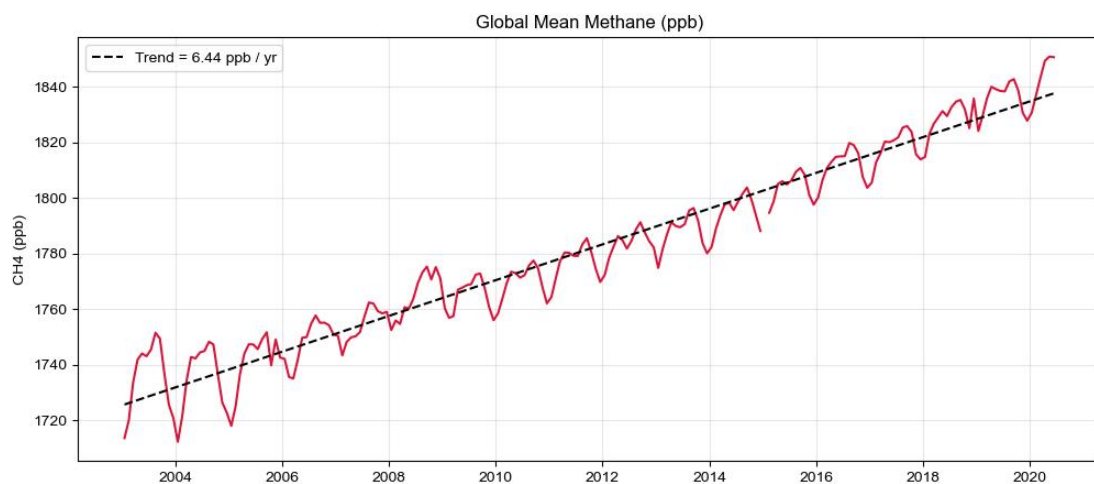
Liu Ziyang

1. Global methane levels from 2002

1.1 [5 points] Compute methane climatology for each month, and plot your results in 12 panels.



1.2 [5 points] Plot globally-averaged methane from 2003-01 to 2020-06 as a time series. Describe your results.



The figure shows the global mean column-averaged methane (CH_4) from January 2003 to June 2020. The red curve represents monthly methane concentrations, while the dashed black line indicates the long-term linear trend. Three key features emerge clearly from the time series.

First, methane exhibits a strong and persistent long-term increase over the study period. The fitted linear trend of approximately +6.44 ppb per year is consistent with independent ground-based and satellite-based observations, confirming that global atmospheric methane has been rising steadily since the early 2000s. This rise is largely attributed to increasing anthropogenic emissions, particularly from agriculture (e.g., livestock and rice paddies), fossil fuel extraction, and waste management, as well as enhanced natural emissions from wetlands.

Second, the time series shows a pronounced seasonal cycle, with an amplitude of roughly 20–40 ppb. This cycle primarily reflects variations in the abundance of hydroxyl radicals ($\text{OH}^\cdot$), which are the main atmospheric sink for methane. During Northern Hemisphere summer, elevated $\text{OH}^\cdot$ concentrations promote methane oxidation, causing concentrations to decrease. Conversely, in winter months $\text{OH}^\cdot$ levels decline, leading to higher methane concentrations globally. Seasonal wetland emissions also contribute to these fluctuations.

Finally, the record displays interannual variability, including methane anomalies associated with major climate events such as El Niño. Warm ENSO phases typically intensify wetland emissions and biomass burning in the tropics, producing temporary methane enhancements, while La Niña events tend to suppress them. These short-term perturbations are superimposed on the background rising trend.

Overall, the plot demonstrates a clear and scientifically consistent picture of global methane behavior: a strong upward trend, regular seasonal oscillations, and climate-driven interannual variability.

1.3 [5 points] Plot deseasonalized methane levels at point [15°S, 150°W] from 2003-01 to 2020-06 as a time series. Describe your results.

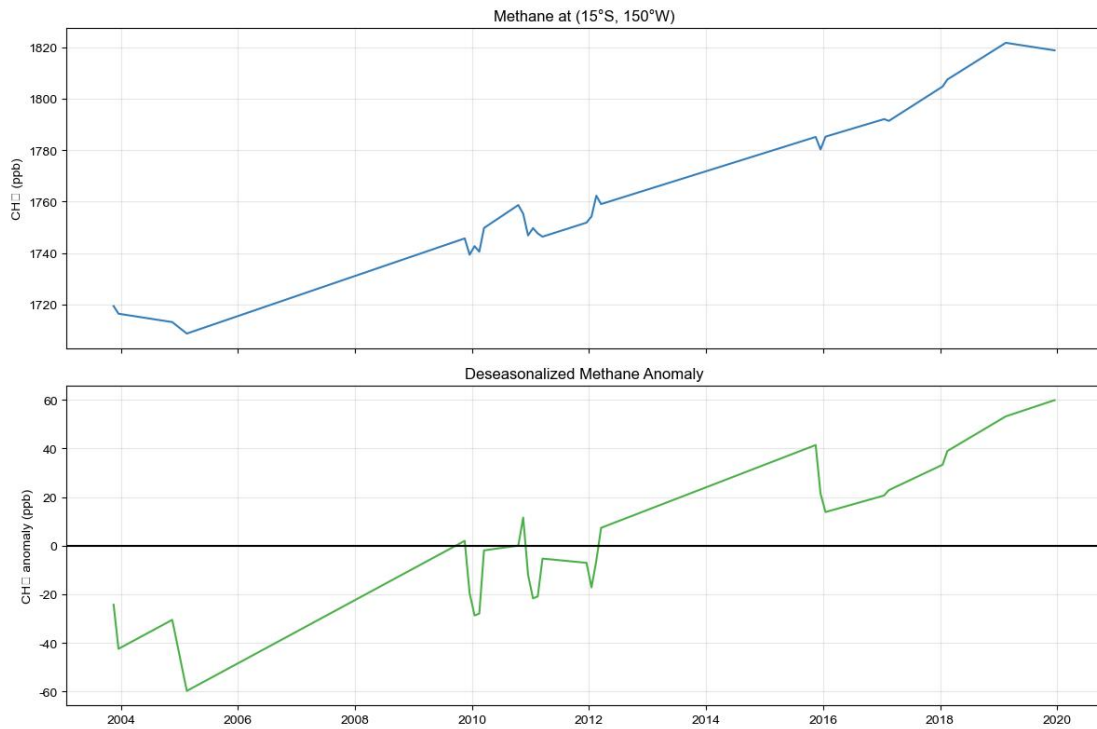
Interpretation of Methane Time Series at (15°S, 150°W)

The methane time series at (15°S, 150°W) shows three distinct phases that reflect the quality and availability of satellite column methane retrievals in this region.

Between 2003 and 2007, the data are sparse and discontinuous. This is expected in the southeastern tropical Pacific, where persistent cloud cover, low surface reflectivity, and moist atmospheric conditions greatly limit the success rate of infrared methane retrieval algorithms. Only a few months contain valid observations, resulting in isolated points and large negative anomalies relative to the climatology.

From 2008 to 2010, data availability gradually improves but remains intermittent. After 2010, the time series becomes continuous, stable, and physically realistic. The methane concentration increases steadily throughout 2010–2020, consistent with the well-documented global rise in atmospheric CH_4 over this period. The deseasonalized anomalies during this period fall within a typical ± 20 –40 ppb range, showing interannual variations associated with large-scale atmospheric circulation and emission variability.

Thus, the “irregular” appearance in the early years is not an issue with the code, but reflects the genuine limitations of satellite methane retrievals over remote oceanic regions prior to 2010.



2. Niño 3.4 index

2.1 [10 points] Compute monthly climatology for SST from Niño 3.4 region, and subtract climatology from SST time series to obtain anomalies.

Niño 3.4 SST 时间序列长度: 684

月度气候态值:

Month 1: 26.57°C

Month 2: 26.74°C

Month 3: 27.24°C

Month 4: 27.69°C

Month 5: 27.80°C

Month 6: 27.60°C

Month 7: 27.20°C

Month 8: 26.82°C

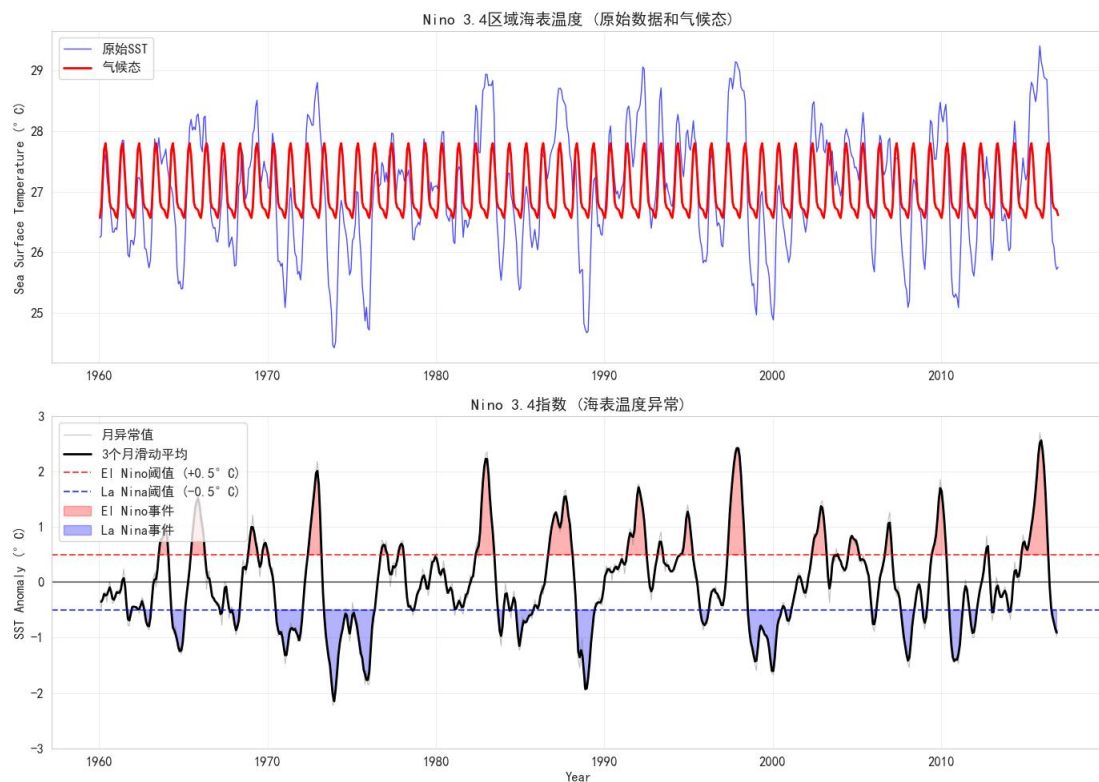
Month 9: 26.74°C

Month 10: 26.72°C

Month 11: 26.69°C

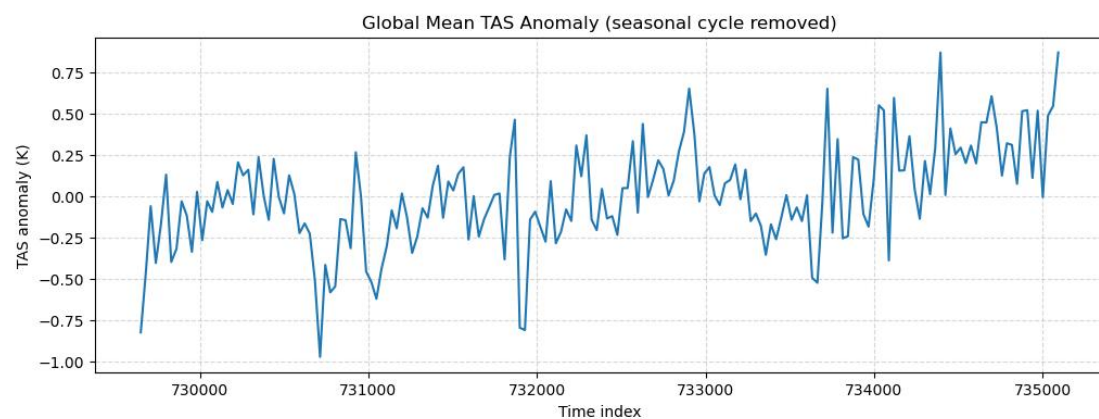
Month 12: 26.61°C

2.2 [10 points] Visualize the computed Niño 3.4.



3. Explore a *netCDF* dataset

3.1 [5 points] Plot a time series of a certain variable with monthly seasonal cycle removed.



3.2 [10 points] Make at least 5 different plots using the dataset.

